

IBD Automation process

Connect Data Science

8 October, 2026

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1 Objective

We created a script to test the current IBDs by performing different checks and tests over the price.

2 Prototype execution:

2.1 Prepare inputs.

The inputs needed are:

- Set of modules for which the scripts will perform the test. We recommend using at least 20 different modules commonly targeted by ibenchmark process (could be provided by IV).
- Tables and chars required to extract from MADRAS the following information related to shops SHOP ID, SHOPTYPE and RETAILER (could be provided by stat ops)
- The script will extract SIRVAL data to perform the tests.

Prepare parameters.

Parameters needed will be:

- SIRVAL and MADRAS connection info.
- **perini & perend** Periods that will be used to extract prices information. It is recommended to use at least two consecutive non-seasonal periods.
- **nankey_percent_change** (default value= 0.15). It will be used to determine which NANKEYS are highlighted as atypical (It takes values from 0 to 1 and the highest the value, the least NANKEYS will be displayed)
- **cv_factor** (default value=2). It collects the times bigger that CV can be compared to the median price to highlight it as atypical.
- **percent_common_assort** (default value=0.3). It collects the minimum percentage of common assortment between shoptypes to perform the TEST.

3 Actions to be taken.

Once established the inputs and parameters, the script will run and perform the test without any required intervention from the user. The output will be CSV file per module containing the SIRVAL-MADRAS raw info (just in case another run is required) and a XLSX file with the results of the tests.

4 Performed tests.

The script will perform tests at different levels:

1. RETAILER-SHOPTYPE level. This test will output a row for each retailer-shoptype checking if the median price per liter/kg/unit falls between the Q25 & Q75 calculated using data from all the other retailers. It will be saved in the sheet “ret_SHOPTYE_outliers” from the excel file.
2. RETAILER-SHOPTYPE-NANKEY level. This test will output the NANKEYS for which the difference between median price for the retailer is compared with the same nankey received on all the other retailers. Cases collected in the sheet “NANKEY_OUT_BOUNDARIES_2” from the excel file will be the ones with a percent of change bigger than **nankey_percent_change**.
3. SHOPTYE level. The test will compare CV from each shoptype (after removing outliers and prices per LT/KG/UN equal or lesser than 0) with the median price per LT/KG/UN for each of the modules. It will output in sheet “SHOPTYE_CV” the cases where CV is **factor** times bigger than the median price per LT/KG/UN.
4. SHOPTYE vs SHOPTYE level. Here a T test is performed in order to prove that average value on one SHOPTYE is equal to the average value for the other shoptypes (comparing 1 vs 1). It will output on sheet “SHOPT vs SHOPT” and column “U_MEANS_P_VALUE2” will take value 1 when means can be considered equal or 0 when they cannot.
5. Complementing the previous output, the excel sheet “ret_no_common_assortment” will collect the pair cases that did not run the previous point (4) because they had not enough common assortment (default is set on 30% using parameter **percent_common_assort**)
6. Finally, there is a sheet named “TEST_NOT_WORKING” where we collect the SHOPTYE pairs for which the TEST did not work. (Mainly it should be empty).

4.1 Evaluate results.

Attending to the excel sheets name we will receive the following outputs.

- ret_SHOPTYE_outliers:

MODULE	RETAILER	SHOPTYE	MEDIAN PRICE	Q25_OTHER_RETAILERS	Q75_OTHER_RETAILERS	STATUS
2801	ES254	<300 ORG	5,2	2,3	5,975	INSIDE BOUNDARIES
2801	ES254	300-799	5,30215	2,39	5,8478	INSIDE BOUNDARIES
2801	ES254	800-1499	5,2917	2,38	5,6042	INSIDE BOUNDARIES
2801	ES223	1500-2499	5,1875	2,0833	5,08	OUTSIDE BOUNDARIES
2801	ES223	2500-6499	5,3333	2,7083	6	INSIDE BOUNDARIES
2801	ES223	300-799	5,375	2,3867	5,8478	INSIDE BOUNDARIES
2801	ES223	800-1499	5,0625	2,38	5,6042	INSIDE BOUNDARIES
2801	ES223	<300 ORG	5,5	2,3	5,9783	INSIDE BOUNDARIES
2801	ES223	GASO	2,0833	2,0833	5,8	INSIDE BOUNDARIES
2801	ES206	<300 ORG	3,78	2,3	5,98	INSIDE BOUNDARIES
2801	ES206	300-799	4,3542	2,39	5,8478	INSIDE BOUNDARIES
2801	ES206	800-1499	4,7231	2,38	5,6042	INSIDE BOUNDARIES
2801	ES322	<300 ORG	5,9	2,3	5,9783	INSIDE BOUNDARIES
2801	ES284	GASO	4,0625	2,0833	5,8125	INSIDE BOUNDARIES
2801	ES041	<300 ORG	5,2708	2,26	5,9	INSIDE BOUNDARIES

To check this sheet, next steps would be to count the number of “OUTSIDE BOUNDARIES” cases compared to the total for each RETAILER-SHOPTYE (by module). Ideal is that the TOTALS are 20 modules, but it can happen that some retailers do not sell products from all of the listed

modules, so we recommend at least a 75% of “Outside Boundaries” cases in order to start a deeper investigation on the possibility of keeping that retailer outside the ibenchmark calculations.

- NANKEY_OUT_BOUNDARIES_2

C_CODEGR	SHOPTYE T	NAN KEY	q25_val T	q75_val T	median_val T	avg_val T	std_val T	unit_stores	cv_val T	AC_CHANNELID	SHOPTYE	q25_val control	q75_val control	median_val control	avg_val control	std_val control	unit_stores control	cv_val control	AC_CHANNELID control
ES102	800-1499	42102518	5,85	7,25	5,85	6,527419355	0,711200711	31	0,108955879	SCAN	800-1499	6,975	8,375	7,625	7,533664524	0,836272334	1711	0,111004722	
ES102	800-1499	45226290	3,5778	4,8333	4,6	4,3886875	0,510083468	56	0,116226883	SCAN	800-1499	3,8333	3,8889	3,8889	3,784693414	0,250451588	2308	0,066174868	
ES102	800-1499	71787146	5,9714	5,9714	5,9714	5,9714	0	6	0	SCAN	800-1499	4,2714	4,2714	4,2714	4,2714	0	3	0	

In this sheet we can find atypical NANKEYs. It is interesting to have a look to these cases just in case any pattern appears (all belonging to same retailer, or focusing in one module, etc).

- SHOPTYE_CV

I do not share any example because we did not receive any positive case on this yet. In any case This info would point to a too wide range of prices for that ibenchmark IBD (shoptye) and it should be considered the option of splitting it.

- U_MEANS_P_VALUE2

MODULE	SHOPTYE	SHOPTYE	UNIT_TAR	UNIT_OTH	ANS_P	V/q25_val T	q75_val T	val cont	val cont	median_val	an_val	coavg_val T	val cont	std_val T	val cont	cv_val T	val cont	ANS_P	VAL
2801	1500-2499	2500-6499	361689	98653	0	2,0833	5,14	2,73	6	3,1	4,6	3,681101	4,631245	1,883878	2,096879	0,51177	0,452768	0	
2801	1500-2499	300-799	361689	643608	0	2,0833	5,14	2,39	5,8478	3,1	4,12	3,681101	4,259615	1,883878	1,942893	0,51177	0,456119	0	
2801	1500-2499	800-1499	361689	655483	0	2,0833	5,14	2,38	5,6042	3,1	3,91	3,681101	4,122803	1,883878	1,944643	0,51177	0,47168	0	
2801	1500-2499	<300 ORG	361689	285418	0	2,0833	5,14	2,3	5,9783	3,1	4,1458	3,681101	4,300343	1,883878	1,988067	0,51177	0,462304	0	
2801	1500-2499	>6500	361689	173716	0	2,0833	5,14	2,7396	6	3,1	4,475	3,681101	4,652511	1,883878	2,150888	0,51177	0,462307	0	
2801	2500-6499	300-799	98653	643608	0	2,73	6	2,39	5,8478	4,6	4,12	4,631245	4,259615	2,096879	1,942893	0,452768	0,456119	0	
2801	2500-6499	800-1499	98653	655483	0	2,73	6	2,38	5,6042	4,6	3,91	4,631245	4,122803	2,096879	1,944643	0,452768	0,47168	0	
2801	2500-6499	<300 ORG	98653	285418	0	2,73	6	2,3	5,9783	4,6	4,1458	4,631245	4,300343	2,096879	1,988067	0,452768	0,462304	0	
2801	2500-6499	>6500	98653	173716	0,011728	2,73	6	2,7396	6	4,6	4,475	4,631245	4,652511	2,096879	2,150888	0,452768	0,462307	0	
2801	300-799	800-1499	643608	655483	0	2,39	5,8478	2,38	5,6042	4,12	3,91	4,259615	4,122803	1,942893	1,944643	0,456119	0,47168	0	

In this case, it has to be checked whether two IBDs (shoptyes) have equal prices taking into account the sum of modules for which shoptyes pairs were considered equal.

- ret_no_common_assortment

MODU	SHOPTYE	SHOPTYE	PERCENT_COMM
2801	1500-2499	NS	0,03
2801	1500-2499	PDORG	0,04
2801	1500-2499	GASO	0,11
2801	2500-6499	NS	0,03
2801	2500-6499	PDORG	0,05
2801	2500-6499	GASO	0,13
2801	NS	PDORG	0,26

Here a list of shoptye pairs for which the condition of at least 30% common assortment was not accomplished and therefore T test was not run. No need to perform any action.